

Presenting Architectural Research in VR

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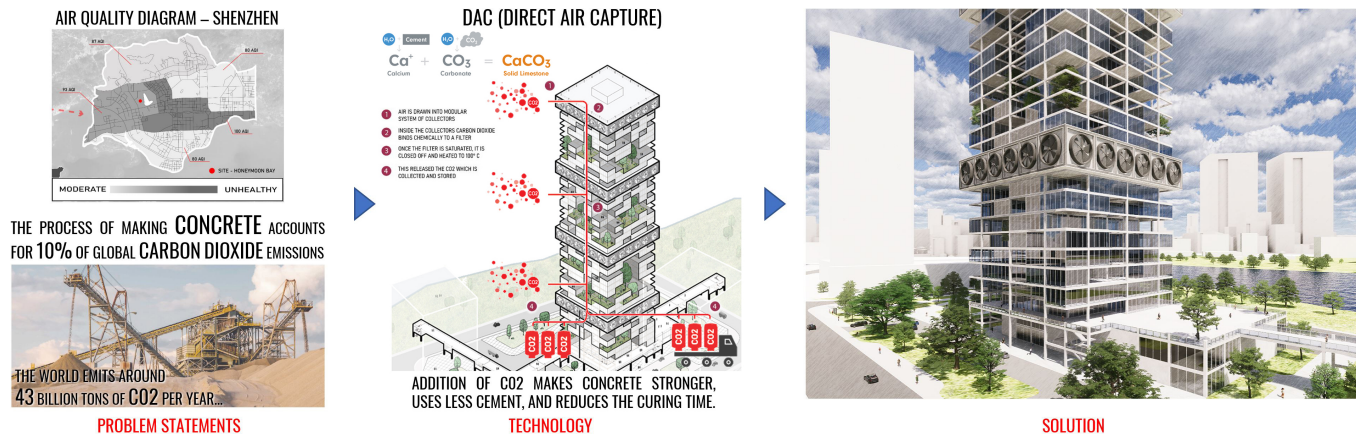


Figure 1: Carbon Tower by Janine Lojko & Mary Riccio. Incorporating direct air capture (DAC) units in the multipurpose skyscraper, as well as including a concrete and carbon facility, this design can clean the air of carbon dioxide while distributing carbon-cured concrete and stored carbon to the surrounding construction and businesses. A VR model for presentation (right).

ABSTRACT

In recent years, new standards for sustainable development and innovations in technologies require us to rethink the forms and functions of architecture that meet emerging needs for future cities. Anticipated developments in digital communication, autonomous vehicles, smart cities, and new energy harvesting technologies are expected to change our lifestyles, leading us to reconsider how we design and organize spatial conditions in our built environment. Such changes will challenge the roles of designers and how they represent and promote their design solutions using new media, including virtual reality (VR) and digital design technologies.

In this submission, I present a design assignment that challenges students to create speculative building designs by applying anticipated technologies that can make our built environment more sustainable. The process in this assignment can be applied to other design fields, including product design and urban design. Many design disciplines can benefit by exploring educational approaches that help integrate new emerging technologies for sustainable product developments and incorporating new digital means for representations for such visions.

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CCS CONCEPTS

• Applied computing → Architecture (buildings); Education.

KEYWORDS

architecture, digital design, VR, technological convergence

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1 INTRODUCTION

In this submission for the Engaging Education Techniques and Assignments session, I introduce a design assignment that challenges students to create speculative building designs by applying anticipated technologies to make our built environment more sustainable. The assignment is project-based and requires creative thinking and multidisciplinary problem solving for speculative scenarios through the interplay of technology and design thinking. The project can be assigned individually or to student groups. The steps for the process of this assignment are summarized as follows. 1). Students are provided with existing site locations, programs for buildings, and any constraints for construction and regulatory codes. 2). Students find existing environmental problems for the proposed site for their building design. 3). Students research both existing and anticipated technology that can resolve the problems. 4). Students integrate the new technology into their architectural solutions creatively. 5). Students explore virtual reality environments to represent their visions to guest experts for assessment and discussion.

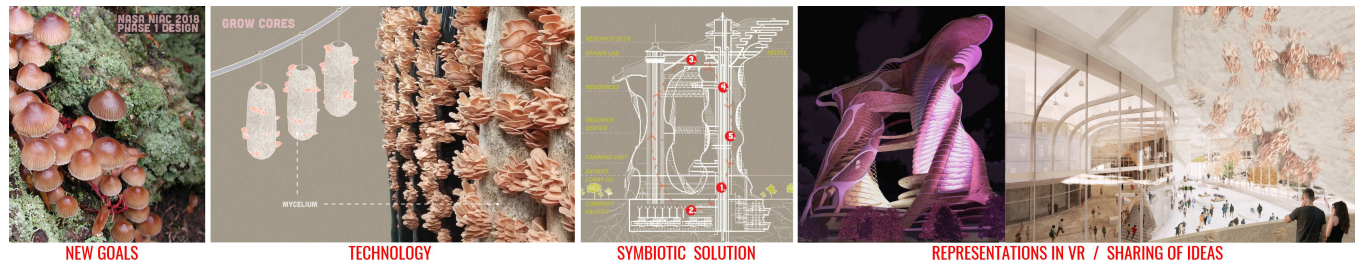


Figure 2: Mycelium Factory by Reyne Bennett (student). Studies by NASA shows applications of fungal mycelium to building constructions using its excellent mechanical properties. The project serves as a metabolic tower designed to vertically cultivate mushrooms for food and research, while facilitating a symbiotic relationship between fungi, bacteria, plants, and humans.

Table 1: Metadata in tabular format

Summary	Students target problems in future living, e.g., pollution, and create design solutions by applying anticipated technologies for sustainability.
Learning outcomes	Students learn and practice creative problem solving discover non-traditional design generation and presentation methods.
Classification	Architecture, VR, technological convergence.
Audience	Undergraduate students at all levels. Goals can be adjusted based on experience and skills.
Dependencies	Varied based on the selected project.
Prerequisites	None. Non-sequential assignment.
Strengths	Scalable in scope and complexity to some other design disciplines. Students learn a non-traditional presentation format including VR. Opportunity to present to experts. Can be assigned individually or to student groups.
Weaknesses	Require students to develop their own goals and requirements in contrast to conventional design exercises based only on goals prepared by instructors, which could be challenging.
Variants	Unlimited design options and approaches.
Assessment	Rubric can be designed around quality and effectiveness metrics. Open discussions with invited experts from industry and academia on presentations using creative communication media.

2 METHODOLOGY

The assignment is designed to motivate creative thinking by challenging students to work within the requirements from the selected site location, climate conditions, purpose, and usage types of buildings, including required areas. During the introduction stage, students study the local conditions, including its culture, climate, industry, and demographics, and target specific existing problem areas for the current environment, such as air quality, the life cycle of building materials currently used, and increasing density. Then, they will seek improvement for future living conditions for the site.

In the next stage of research, students search for technology and innovative strategies to remedy the problems found from the current site conditions in the previous step. Students learn from research papers and articles that explain scientific mechanisms to improve the underlining problems in the cities and existing application examples of the technology from scientific databases (ACM, IEEE) using search engines (Google Scholar, etc.).

After the research stage, students find creative solutions to incorporate the selected technology into their architectural designs. Starting with diagrammatic floor plans representing adjacency among different rooms for commonly used building types, such as office or residential towers, students fit the proposed technology and their architectural strategies into the preliminary layouts incorporating emerging new programs associated with the technology. For example, some groups introduced the visitor center and demonstration facilities for the adopted new technology.

After students' designs get matured, they find the best way to represent their architectural visions to hypothetical stakeholders, clients, investors, and collaborators such as engineers who could technically support their projects. Tutorials on presentations using immersive VR in a computer game development environment (Twinmotion, Unity3D, Unreal, etc.) were introduced by the instructor, which helped students develop their own creative approaches to articulate each project's strength (see Table 1 for details).

Upon completion, media can be reviewed by invited experts from industry and academia in class. Reviewers rated students' VR, drawings, and slide presentations based on concept, aesthetics, and technical accomplishments. Contents in immersive VR are used not only for representations but also for evaluations of spaces yet to be built inside the simulated environments through the lens of virtual. Comments from reviewers promote discussions as to how students should further proceed with their explorations.

3 CONCLUSION

To become the next leader in the field of design, students need to have the habit of developing one's creative solutions to meet the new standards. Simply following templates and receipts from existing design solutions might no longer be sufficient to provide innovative solutions that can respond to ever-changing dynamic demands from society under the era of technological convergence. This project aims to prepare students for such future scenarios as they venture into the professional world.